# Blessing O. Okosun

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## Education

**Doctor of Philosophy - Biology** *Dec. 2026*
*University of North Dakota, Grand Forks, ND*
Advisor: Diane Darland, Ph.D.
Dissertation: *Vascular-derived factors and their influence on the developing mouse cortex*

**Communicating Science Graduate Certificate** *May 2025*
*University of North Dakota, Grand Forks, ND*

**Micro certification - Mentors Helping Mentors** *Feb. 2024*
*University of North Dakota, Grand Forks, ND*

**Bachelor of Science in Biology** *May 2021*
*Dickinson State University, Dickinson, ND*
Major: *Cell and Molecular Biology*
Minor: *Chemistry*
Honors Thesis: *Arabidopsis Responsiveness to N, N-dimethylheadecylamine*

**Bachelor of Science in Science Laboratory Technology** *Dec. 2015*
*University of Benin, Benin City, Nigeria*
Major: *Microbiology*
Honors Thesis: *Effects of Antimalarial Drugs on Blood Transfused Patients in University of Benin Teaching Hospital, Edo State*

**Nigerian Certificate of Education** *March 2009*
*College of Education, Ekiadolor, Nigeria*

## Scholarly Tools

**Bioinformatics**
*Specialized in Single-cell & RNA-Seq analysis*

**Advanced Imaging**
*Specialized in the use of different confocal microscopy methodologies and imaging analysis*

## Research Experience

**Graduate Research Assistant** *Summer Semester 2021 - Present*
*University of North Dakota, Grand Forks, ND*
*Supervisor: Dr. Diane Darland*

- 2025; Using flow cytometric analysis, immunolabeling, single-cell analysis, and confocal microscopy to determine the role of leptin/leptin receptors in determining neural stem cell fate. I also worked with a collaborator on the bioapplication of nanozymes for detecting hydrogen peroxide release from metabolically stressed endothelial cells.

- 2024; Used phylogenetic analysis to determine the evolutionary relationship of leptin receptors across different members of Deuterostomes. I also worked with a collaborator on the bioapplication of nanozymes for the detection of reactive oxygen species in metabolically stressed endothelial cells.
- 2022 - 2023; I utilized co-culture and organoid models, along with bulk RNA sequencing, PCR, and RT-qPCR techniques, to investigate how the differential distribution of vascular endothelial growth factor isoforms affects the expansion of the embryonic mouse cortex. Additionally, I focused on the bioapplication of polymer-based nanoparticles carrying siRNA to knock down the expression of selected genes in brain-derived microvascular endothelial cells and fibroblasts. Furthermore, I explored the bioapplication of persistent luminescence polymer-based nanoparticles in brain-derived microvascular cells.
- 2021; used magnetic beads and antibodies to select for neural stem cells from mice embryos' brains and generate clonal cell lines.

**IDeA Networks of Biomedical Research Excellence Research Assistant** *Sept. 2019 - Jan. 2021*
*Dickinson State University, Dickinson, ND*
*Supervisor: Dr. Craig Whippo*

- Determine different Arabidopsis responsiveness to volatile organic compounds. This was done by characterization of the phenotypic natural variations of the different parts of imaged Arabidopsis that were grown in media. Techniques employed were gas sterilization, image analysis on ImageJ, and data processing in R.
- Carried out a Genome wide Association Study to determine the genetic variation and diversity of 198 different Arabidopsis accessions.

**NSF Research Experiences for Undergraduates** *May. 2019 - Aug. 2021*
*South Dakota School of Mines & Technology*
*Supervisor: Dr. Timothy Brenza & Jordan A. Hoops*

- Characterization of Calu-3 cell line culture protocol by determining if air-liquid interface method (ALI; that mimics cell conditions in the lungs) is better for drug delivery compared to the traditional method of submerging cells in media.
- Recording and documenting transepithelial electrical resistance, permeability of fluorescence molecules to mimic drug delivery and growth kinetics quantified by MTT viability assay (to measure cell viability) and cell counts.

**Laboratory Technologist** *Jan. 2014 - Dec. 2015*
*Russel International Group of Schools, Benin City, Nigeria*

- Maintaining safety and inventory for different STEM labs within the school as well as ordering supplies.
- Preparing samples and setting up different experiments for each lab at he recommendation of the subject head teacher.

**Industry-based Internship** *Feb. 2013 - June 2013*
*Oil Mill Analytical Laboratory, Okomu Oil Palm Company, Edo state, Nigeria*
*Supervisor: Engr. Omokaro Osaikhuiwuomwan*

- Collection and determination of the free fatty acid in samples collected at different stages on the production of crude palm oil, palm kernel oil and palm kernel cake.

## Teaching Experience

**Graduate Teaching Assistant** *Fall Semester 2022 - Present*
*Department of Biology University of North Dakota, Grand Forks, ND*

- 2022 - 2025 taught as the sole instructor General biology laboratory under the assistant of a faculty member. At the same time, each year, I taught Genetics class as a co-instructor where I assisted with case studies and implementation of other active learning activities.

**Graduate Teaching Assistant** *Spring Semester, 2023 - Present*
*Department of Biology University of North Dakota, Grand Forks, ND*

- 2023 - 2024, 2026 taught Cell biology labs as the sole instructor under the guidance of a faculty member.
- 2025 taught Cell biology class as a co-instructor.

**Middle School Instructor** *2010 - 2012*
*Ekpon Junior Secondary School, Edo State*
*Universal Basic Education Scheme, Nigeria*

- Taught Integrated Science class to three class ages in the school from October 2010 to September 2012.

**Elementary School Instructor** *2009 - 2010*
*Bedrock Education Center, Benin City, Nigeria*

- Taught as head teacher of an elementary school class from August 2009 to June 2010.

## Professional Development

**Workshop Participant, Having Difficult Conversations** *Dec. 13th, 2025*
*University of North Dakota*

**Attendee, 3rd Structural Birth Defects Trainee Symposium** *Sept. 23-25th 2025*
*Virtual meeting, Society of Developmental Biology*

**Workshop Participant, Effective Communication** *June 18th, 2025*
*University of North Dakota*

**Trainee, Graduate Teaching Assistants' Best Practices for College Teaching** *Aug. 30th, 2024*
*University of North Dakota*

**Trainee, Virtual 3-Day Single-cell RNA-Seq Analysis** *June 6-9th, 2025*
*Institute for Quantitative & Computational Bioscience, UCLA, CA*

# Grants

**Academic Programs & Student Awards | $250** *Oct. 2025 - May 2026*
*Department of Biology, University of North Dakota*
Project goal: *Investigation of the differential expression of leptin receptor in the forebrain on embryonic mice.*

**Esther Wheeler Wadsworth Hall Award | $1400** *May 2024 - Dec. 2024*
*Department of Biology, University of North Dakota*
*Project goal*: Investigation of the differential expression of leptin receptor in neural stem cells treated with different growth factors.

**Research Assistant Award | $6500** *May 2023 - Dec. 2023*
*Department of Biology, University of North Dakota*
*Project goal*: Co-developer of a micro certification program for the University of North Dakota. This was under the 2023 summer professorship project titled Strengthening the Future of STEMM: Mentors Helping Mentors.

**Academic Programs & Student Awards | $480** *Oct. 2023 - May 2024*
*Department of Biology, University of North Dakota*
*Project goal*: To design and purchase different leptin receptor primers for a, b, and c isoforms.

**Research Assistant Award | $1111** *May 16 - 31st 2023*
*New Discoveries in the Advance Interface of Computation, Engineering & Science, NSF EPSCoR*
*Project goal*: To develop a cell-to-tissue size ratio model that K2 to 12 STEM teachers in North Dakota Elementary schools will use for teaching. The products were a talk, a poster presentation, and an exhibition of the model.

**Doctoral Research Grant | $500** *June 2023 - Aug. 2023*
*School of Graduate Studies, University of North Dakota*
*Project goal*: To attend a 3-day hands-on virtual workshop on single-cell analysis on R.

**Doctoral Research Grant | $500** *June 2022 - Aug. 2022*
*School of Graduate Studies, University of North Dakota*
*Project goal*: To cover the cost of confocal microscopy imaging of different forebrain sections in embryonic mice expressing various or single vascular endothelial growth factor isoforms.

# Awards and Honors

- 1st place, Graduate Oral Presentation *April 18th, 2026*
  *North Dakota Academy of Science 117th Annual Meeting*
- 3rd place, NDAS Intersection of Science & Art competition *April 18th, 2026*
  *North Dakota Academy of Science 117th Annual Meeting*
- Earwicker Award for Excellence in Research & Science Communication *April 5th, 2025*
  *North Dakota Academy of Science 116th Annual Meeting*
- Gates Institute Travel Award *March 19th, 2024*
  *Southwest Regional Meeting for the Society for Developmental Biology*

- Best Poster Award *March 28th, 2024*
  *Northern Plains Biological Symposium, University of North Dakota, Grand Forks*
- Dean's list *May 27th, 2020*
  *Dickinson State University, Dickinson, North Dakota*
- Dean's list *Jan. 23rd, 2020*
  *Dickinson State University, Dickinson, North Dakota*
- Dean's list *Jan. 7th, 2019*
  *Dickinson State University, Dickinson, North Dakota*

# Publications

**Peer reviewed**

- Wu, Y., Combs, C., **Okosun, B. O.,** Tayutivutikul, K., Darland, D. C., & Zhao, J. X. (2025). Fe3+-Doped Graphene Quantum Dots-Based Nanozyme for H2O2 Detection in Cellular Metabolic Distress. ACS Applied Nano Materials. https://doi.org/10.1021/acsanm.4c06292.
- Sun, D., **Okosun, B. O.,** Xue, Y., Tayutivutikul, K., Smith, K. H., Darland, D. C., & Zhao, J. X. (2025). Multi-functional near-infrared fluorescent polymer dot–siRNA for gene expression regulation. Journal of Materials Chemistry B, 13(6). https://doi.org/10.1039/D4TB01954G.
- Darland, D.C., Gisi, E.M., Hampton, J.R., Huang, H., Kantonen, L.M., Kong, D., *LaFond, L.R., Martin, J.P., **Okosun, B.O.**, Westberg, L.M., and Young, L.C. (2024). Student-centered approaches to breaking through scientific writing barriers. Journal of College Science Teaching. https://doi.org/10.1080/0047231X.2024.2373022. Authorship order is alphabetical by last name.

**In Review**

- **Okosun, B. O.,** Mohammed, K. A., Owolabi, P. J., Khalid, A. S., Abusaad, H. E., Anifowose, A. S., & Alfaki, M. (2026). Pan cancer analysis reveals context dependent prognostic roles of PTEN across human cancers. Submitted to Discover Oncology.
- Che-Mughe, H., **Okosun, B.O.,** Ozdogan, M., Kola, H.A., Oncel, N., Darland, D.C., and Zhao, J.X. (2026) Development of Iron and Nitrogen Co-doped Graphene Quantum Dots-based Nanozyme for Detection of NADPH. Submitted to ACS Applied Nano Materials.

**In Preparation**

- **Okosun, B.O,** Smith K.H., Tayutivutikul K., Kempel K.K, Di, S., Zhao, J.X. and Darland, D.C. Leptin Receptor and its novel role in cortical development., in preparation, 2026.
- **Okosun, B.O.,** Smith K.H., Tayutivutikul K., Kempel K.K, Di, S., Zhao, J.X. and Darland, D.C. Leptin Receptor at the neurovascular interface in cortical development, in preparation, 2026.

**Public Outreach & Popular Press**

- **Okosun, B.** (2025). An unplanned pregnancy rocked my Ph.D. My academic village helped me make it through. Science. https://doi.org/10.1126/science.zxqs65q.

- Training grad students to be effective mentors, by Walter Criswell. Published in: Affinity, School of Graduate Studies, TTaDA, Nov. 14th 2024.

- Science Student of the Week, Department of Natural Science. Published in: Dickinson State University News. March 18th, 2021.

# Presentations (* indicates presenter)

**Talks**

- From metabolism to vasculature: leptin effects on brain microvascular endothelial cells. **Blessing O. Okosun***, Kaylee K. Kempel, and Diane C. Darland. North Dakota Academy of Science; 117th Annual Meeting, Memorial Union, North Dakota State University, Fargo, ND. 5th April, 2025.

- Beyond appetite—Leptin orchestrates critical neural stem cell differentiation. **Blessing O. Okosun***, Kirati Tayutivutikul, Kaylee K. Kempel, Kaitlyn H. Smith, and Diane C. Darland. North Dakota Academy of Science; 116th Annual Meeting, Memorial Union, University of North Dakota, Grand Forks, 5th April, 2025.

- Science, Technology, Engineering, and Mathematics—Module. *Zhao, J.X.,* Wu, Y., *Sun, D.,* ***Okosun, B.O***., Xue, Y.*, Baird, J., Summers, R., and Darland, D.C*. Nurturing STEM in the PreK-12 Classroom (Oral Presentation), Memorial Union, University of North Dakota, Grand Forks, August 10, 2023. *Indicates Author, Presenter and Equal Contributor.

- Science and Scale from Atoms to Ecosystems. *Zhao, J.X.,* Wu, Y., *Sun, D.,* ***Okosun, B***., *Xue, Y., Baird, J., Summers, R., and* Darland, D.C. Nurturing STEM in the PreK-12 Classroom Conference (Exhibition and Demonstration), Memorial Union, UND, Grand Forks, August 10, 2023. *Indicates Author, Presenter and Equal Contributor.

- Vascular Cells and their influence on the neural stem cell microenvironment of the Cortex. Kirati Tayutivutikul*, **Blessing O. Okosun**, and Diane C. Darland. North Dakota Academy of Science; 114th Annual Meeting, University of North Dakota, Grand Forks, ND. April 5th, 2023.

- Dual Functional near-infrared Fluorescent Polymer Dots-siRNA for Dynamic Delivery and Detection. Di Sun*, **Blessing O. Okosun**, Kirati Tayutivutikul, Diane C. Darland, and Julia X. Zhao. North Dakota Academy of Science; 114th Annual Meeting, University of North Dakota, Grand Forks, ND. April 15th, 2023.

- Arabidopsis Responsiveness to N, N-dimethylhexadecylamine. **Blessing Okosun***, and Craig Whippo. Celebration of Scholars, Dickinson State University, Dickinson, North Dakota, April 19th, 2021.

- Characterization of Calu-3. **Blessing Okosun***, Jordan A. Hoops, Timothy M. Brenz. South Dakota Undergraduate Research Symposium & Faculty Meetings, Sioux Falls, South Dakota, July 30th, 2019.

**Posters**

- From metabolism to vasculature: Using omics approaches to link Leptin receptor (LepR) to novel functions in vascular-related diseases. **Blessing O. Okosun***, Kaylee K. Kempel, and Diane C. Darland. The festival of Genomics, Biodata & AI, Boston. June 3rd - 4th, 2026.

- Leptin as a Pluripotent hormone—Cortical neural stem cells display differential responses to the leptin hormone. **Blessing O. Okosun***, Kirati Tayutivutikul, Kaylee K. Kempel, Kaitlyn H. Smith, and Diane C. Darland. Graduate Research Achievement Day, University of North Dakota, February 27th, 2025.

- Environment matters - Cortical Neural Stem Cells Display Differential Response to Vascular-derived Factor. **Okosun, B. O.***, Tayutivutikul, K., Kempel, K., and Darland, D.C. Society for Developmental Biology Southwest Regional Meeting, University of Colorado Anschutz Medical Campus, Denver, CO, March 4th, 2024.

- Leptin Receptor signaling in cortical Development. Smith, K.H., ***Okosun, B.O.,*** Tayutivutikul, K., Tkach, E., and Darland, D.C. Society for Developmental Biology, Southwest Regional Meeting, University of Colorado Anschutz Medical Campus, Denver, CO, March 3rd, 2024.

- Environment matters - Cortical Neural Stem Cells Display Differential Response to Vascular-derived Factor. **Okosun, B. O.***, Tayutivutikul, K., Kempel, K., and Darland, D.C. Northern Plains Biological Symposium, University of North Dakota, Grand Forks, March 28th, 2024.

- Multi-functional Near-infrared Fluorescent Polymer Dot (Pdot)-siRNA for Gene Expression Regulation. Di Sun, ***Blessing O. Okosun***, Kirati Tayutivutikul, Kaitlyn H. Smith, Diane C. Darland, and Julia X. Zhao. Epigenetics & Epigenomics Symposium, University of North Dakota, May 21st, 2024.

- Pick your Poison: How Aspartame Causes Anxiety. Abby J. Kersey, ***Blessing O. Okosun***, Baylee Kram, Diane C. Darland. UNDergraduate Showcase, University of North Dakota, Dec 7th, 2024.

- Cell & Tissues. Science and Scale from Atoms to Ecosystems. **Okosun, B.***, and Darland, D.C. Nurturing STEM in the PreK-12 Classroom Conference, Memorial Union, UND, Grand Forks, August 10, 2023.

- Dual Functional Near-infrared Fluorescent Polymer Dots-siRNA Nanomedicine for Cancer Gene Therapy. Di Sun, ***Blessing Okosun***, Diane Darland, and Julia Zhao. Epigenetics & Epigenomics Symposium, University of North Dakota, May 19th, 2023.

- Leptin Regulation and Metabolism: Putting the Focus on ADHD. Carter T. Krenelka, ***Blessing O. Okosun***, Diane C. Darland. UNDergraduate Showcase, University of North Dakota, May 3rd, 2023.

- The dual regulatory role of vascular endothelial growth factor (Vegf) in neural and vascular development in the Mouse Cortex. **Blessing O. Okosun***, Gabel, D. R., Tkach, E, and Diane C. Darland. Graduate Research Achievement Day, University of North Dakota, March 3rd, 2022.

- Arabidopsis Responsiveness to N, N-dimethylhexadecylamine. **Blessing Okosun***, Craig Whippo. Celebration of Scholars, Dickinson State University, Dickinson, ND April 19th, 2021.

- Characterization of human lung adenocarcinoma cell line Calu3 under air-liquid interface culture. **Blessing Okosun***, Jordan A. Hoops, Timothy M. Brenz. South Dakota Undergraduate Research Symposium & Faculty Meetings, Sioux Falls, SD, July 30th, 2019.

# Science Communications and Community Outreach

- Judge *2023 - Present*
  - North Dakota Science and Engineering Fair.
  - Judging projects under Life Sciences competition at the Regional and State level. The projects were from seventh- through-twelfth grade students. In 2025, I served as a Chair of Judges of a session.

- Intern *May – Aug 2025*
  - InternGF Program, Grand Forks Region Economic Development Corporation.
  - Selected participant in a Grand Forks professional development program connecting graduate researchers with local innovation, entrepreneurship, and community outreach initiatives.

- Interviewee *XX*
  - High School STEM Documentary Project.
  - Featured in a student-led video interview initiative highlighting ongoing neuroscience research and graduate school experience.

- Pen pal *2024 - Present*
  - Letters to a Pre-Scientist Program Volunteer Science pen pal.
  - I write to middle school students across the U.S. to foster STEM curiosity and a sense of science identity. Shared personal scientific experiences and answered student questions in an accessible language. Aimed at increasing visibility of STEM pathways for younger audiences.

- Co-developer and participant *2023 - 2024*
  - UND Graduate School Summer Professorship.
  - Co-developed a peer-mentoring program called the Mentors Helping Mentors program for graduate students at the University of North Dakota. I also participated as a student in the first session of the program.

- Coordinator of session *2019 - 2021*
  - Science Olympiad, Dickinson State University, ND.
  - Planning, coordinating, and judging events under the Life Sciences competition at the regional level.

## Professional Affiliations

- Member | Grand Forks Young Professionals *2025 - Present*
- Treasurer | Nigerian Students Association, University of North Dakota *2025 - Present*
- Member | Epigenetics & Epigenomics Group *2023 - Present*
- Member | Society for Developmental Biology *2021 - Present*
- Member | National Center for Faculty Development & Diversity *2021 - Present*
- Treasurer | Science Club, Dickinson State University, North Dakota *2020 - 2021*
- Associate | Nigerian Institute of Science Laboratory Technology *2016 - 2018*

## Statements (full documents available on request)

**Research Statement, in Brief.**
My research interests are centered on understanding the co-dependence mechanism underlying congenital neuro-malformations. To achieve this, my work currently focuses on how the development of the neural and vascular systems is coordinated during early mouse cortical development. I am focused on three critical developmental time points: E9.5, when the cortex is a simple neuroepithelial layer; E11.5, when there is increased proliferation and the onset of neural stem cell differentiation, with early ingression of blood vessels; and E13.5, when there is early cortical layering with extensive vascular investment. These early time points lay the foundation for cellular expansion, neurogenesis, gliogenesis, and cortical function. My specific research goal is to investigate the dual regulatory roles of the Vascular Endothelial Growth Factor (and its protein isoforms) and the leptin hormone in neurovascular development in the embryonic mouse cortex. The results of this research will expand our understanding of the integrated development of the neural and vascular systems with implications for the cortical disruption that can lead to pathology.

**Mentoring Philosophy in Brief.**
Mentoring is like a two-way street of effort, work, and communication between the mentee and the mentor. The overall goal of my mentoring relationship is not to make a professional clone of myself but to help my mentee identify their strengths, turn their weakness into strengths, make great strides in achieving their goals, and prepare them for their next professional journey. By aligning expectations, continuous assessment, effective communication, equity and inclusion, fostering independence, and promoting professional development, I aim to create a mentoring environment that inspires growth, resilience, and success.

- Undergraduate Mentoring through Research Experiences. Over the last three years, I have mentored three undergraduate students, working with them on various aspects of my research. These students are included as co-authors on several of my presentations and will be (or are) co-authors on manuscripts with me.
- Undergraduate Mentoring through Team Presentations. In the Fall 2023 Neuroscience and Nutrition Seminar Course, I mentored two undergraduate students doing a Team presentation (BIOL491/503). This involved facilitating research topic and article selection, developing a presentation slide deck, coordinating contributions to the presentation, and managing formal presentation/discussion.